

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I Jagadeil (192521354) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. J. S.

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services.
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CSA15 - cloud computing and Big Data Analysis

ASSESSMENT TOOL - I

Short Answer Test!

1) The evolution computing isolated personal powerful based system computing over internet development efficiency.

a) Standalone computing,

- * each computer worked independently
- * no resource sharing

b) Client - server computing!

- * Central servers store data and applications
- * Improved data management.

2)

A hypervisor is software that enable multiple virtual machines (VMS) to run a single physical computer by sharing hardware resources.

Type 1 Hypervisor (Bare metal)

- * Runs directly on hardware
- * No host operation system
- * High performance and security

Type 2 Hypervisor (Hosted)

- * Runs on top of an existing
- * easier to install and use
- * Suitable for personal testing

3)

An application programming interface (API) is a set of rules that allows different software applications to communicate or exchange data.

* Role of APIs

- connect cloud applications
- Enable data exchange
- Automate cloud operations

* APIs support interoperability

- Allowing communication
- provide standard
- Integrating SaaS, PaaS and IaaS.

4) Resource pooling in cloud computing

Resource to cloud characteristic where computing resources among multiple users using a multi-tenant model.

Cloud providers maintain a pool of.

- * servers
- * storage
- * networks
- * memory
- * processing power.

5)

Rapid elasticity ability cloud system automatically increase or decrease computing resources based on workload demand.

- * Resource scale up during high demand
- * Resource scale down when demand decreases
- * Scaling occurs automatically without delay.